

电路理论 mooc 答案——第十二章

解析

1.解:

共4个电感、3个电容。

$$KVL: u_{C1} + u_{C2} + u_{C3} = 0$$

$$KCL: i_{L3} + i_{L4} + i_{S2} = 0$$

$$KCL: i_{L1} + i_{L2} + i_{L3} + i_{L4} + i_{S2} = 0 \text{ (中间整体作为黑箱, 有个高斯面)}$$

$$\text{复杂度} = 7 - 3 = 4$$

选B

2.选D

Remark: 不会就去看书

3.选A

4.解:

$$A. u_+ = u_- = u_S = \epsilon(t) \Rightarrow i_{R1} = \frac{u_+}{R_1} = \text{const if } t > 0.$$

$$B. KCL: i_{R1} = \frac{u_o - u_-}{R_3} + C \frac{d(u_o - u_-)}{dt} \Rightarrow \tau = R_3 C$$

$$C. i_{total} = i_{R1} = \text{const}$$

$$D. u_o = u_S + \frac{R_3}{R_1} u_S \left(1 - e^{-\frac{1}{\tau} t} \right), \text{ then } u_o(0^+) = u_S = 1V$$

选B

5.解:

$$i_L(0^-) = 0A, u_C(0^-) = 10 * \frac{3}{2+3} V = 6V$$

$$\text{Then, } i(0^+) = i_L(0^-) = 0A, u_C(0^+) = u_C(0^-) = 6V, \left. \frac{du_C}{dt} \right|_{0^+} = \frac{1}{C} i(0^+) = 0V/s$$

选C

6.解:

$$i_L(0^-) = \frac{1}{2} A, u_C(0^-) = 10V. (\text{实际上毫无作用})$$

$$i_C = C \frac{du_C}{dt} = -i_L \Rightarrow u_L = L \frac{di_L}{dt} = -LC \frac{d^2 u_C}{dt^2}$$

$$KVL: u_C + i_C R_C = u_L + i_L R_L \Rightarrow LC \frac{d^2 u_C}{dt^2} + (R_C + R_L) C \frac{du_C}{dt} + u_C = 0$$

$$\Delta = (RC)^2 - 4LC > 0 \Rightarrow \text{欠阻尼}$$

选C

Remark: 可见, 电路初态储能并无实际作用。

本题实际上可以给右侧回路一个 $u_s = \epsilon(t)$, 并假设电感电容无储能, 用运算法计算回路电流的复频域表达式, 观察特征根分布判断。

7.解: 本质上就是列基尔霍夫方程的题, 选B.

8.解:

Zero-state response:

$$H(s) = \frac{U_C}{U_s} = \left(\frac{1}{s} - \frac{1}{s+1} \right) \cdot s = \frac{1}{s+1}$$

$$\text{If } u_s(t) = 2\delta(t), U_C = H(s)U_s(s) = 2 \cdot \frac{1}{s+1} \Rightarrow u_C = 2e^{-t}V$$

Zero-input response:

$$\text{unit step response } u_C(t) = (1 - e^{-t})V \Rightarrow \tau = R_{eq}C = 1s$$

$$\text{Therefore, if } u_C(0^-) = 3V, u_C(t) = 3e^{-t}V.$$

$$\text{Complete response} = \text{zero-state response} + \text{zero-input response}, u_C(t) = 2e^{-t} + 3e^{-t}V = 5e^{-t}V$$

选A

Remark: 题目给的 $u_R(t)$ 确实没啥用, 别问了辅导员也很疑惑。

9.解:

$$u_{C2}(0^-) = 2V.$$

$$Y_{eq} = sC_1 + 2G = 0.5s + 1 \Rightarrow Z_{eq} = \frac{2}{s+2}$$

$$I_{SC} = \frac{2}{s} + (0.5s) \cdot \frac{8}{s} = 4 + \frac{2}{s} \Rightarrow U_{OC} = Z_{eq} I_{SC} = \frac{2(4s+2)}{s(s+2)} = \frac{2}{s} + \frac{6}{s+2}$$

$$KVL: U_{OC} = U_{C2} + Z_{eq} \cdot C_2 [sU_{C2} - u_{C2}(0^-)] \Rightarrow U_{C2} = \frac{U_{OC} + Z_{eq} \cdot C_2 u_{C2}(0^-)}{1 + Z_{eq} \cdot (sC_2)} = \frac{\frac{2}{s} + \frac{6}{s+2} + \frac{2}{s+2}}{1 + \frac{2}{s+2} \cdot (0.5s)} = \frac{3}{s+1} + \frac{2}{s} \Rightarrow u_{C2} = (2 + 3e^{-t})\epsilon(t)V$$

选C

10.解:

$$\text{改接前, } Z(s) = 2 + \frac{2}{s}$$

$$H(s) = \frac{U(s)}{U_s(s)} = \left(-\frac{0.1}{s+0.5} + \frac{0.6}{s+3} \right) \cdot (s+3) = \frac{s}{2s+1} = \frac{1}{1 + \left(1 + \frac{1}{s}\right)} = \frac{2}{Z+2}$$

$$\text{改接后, } Z(s) = 1 + 0.5s, \text{ then } H(s) = \frac{2}{Z+2} = \frac{4}{s+6} \Rightarrow U(s) = H(s)U_s(s) = \frac{8}{s+6}$$

选B

Remark: 可以这么分解的原因是, 除源以外, 复频域电路中所有的 s 分量仅可由 RC 、 RL 串联支路提供 (仅 LC 会出现 s 分量)。